NAVNEET RAGHAV

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EDUCATION

National Forensic Sciences University — Lucknow, India
B.Tech in Computer Science and Engineering (Cybersecurity) — 2024 – Present
Cumulative GPA: 8.79/10

Army Public School — Agra, India
Senior Secondary (CBSE) — 2023
Grade: 84.4%
Secondary (CBSE) — 2021
Grade: 92.6%

EXPERIENCE

Research Intern — Varanasi, India
Indian Institute of Technology (IIT BHU), Varanasi — May 2026 – Present
Supervisor:Prof. Harsh Kasyap

- Working on ML privacy and machine unlearning, with a focus on using Membership Inference Attack-based audits to check whether small forget sets still leave detectable traces after approximate unlearning.
- Developed and tested TRACE-UMIA, an internal representation-based audit method that compares token-level hidden-state drift before and after unlearning in transformer text classifiers.
- Ran experiments on DistilBERT and DistilRoBERTa using AG News, QNLI, and SST-2, along with output baselines, TC-style baselines, exact-retraining controls, and utility checks.

Summer Intern — Warangal, India
National Institute of Technology, Warangal (NIT Warangal) — June 2026 – Present

- Working on smart energy systems with a focus on AI-ML, IoT-based sensing, and intelligent monitoring for energy-efficient applications.
- Developing an energy-efficient intrusion detection system for industrial use-cases, funded by Hitachi Energy India Ltd.
- Developed an XGBoost-based ML pipeline, on a custom dataset to detect human heat signatures using a thermal array sensor.

PROJECTS

TRACE-MultiStat: Internal Drift Audit for Small-Set Unlearning — GitHub | Python, PyTorch, Transformers

- Built a PyTorch based audit pipeline to study whether small forget sets still leave internal traces after approximate unlearning.
- Extracted token-level hidden state drift before and after unlearning, and used simple multi-statistic features for forget-vs-retain and forget-vs-unseen tri class auditing.
- Compared TRACE-MultiStat with output-only and TC-UMIA method style baselines using DistilBERT and DistilRoBERTa on AG News, QNLI, and SST-2.

Deepfake Detection using CNN Ensemble and Grad-CAM — GitHub | Python, PyTorch, OpenCV, Streamlit

- Developed a deepfake detection system using XceptionNet and EfficientNet-B0 with ensemble learning, achieving around 90% accuracy.
- Added Grad-CAM visualizations to highlight which parts of the face the model focused on, helping interpret predictions and debug failure cases.
- Handled class imbalance using weighted sampling and loss functions, and added augmentations like JPEG compression, noise, and blur to improve performance on real world images.
- Deployed the system as a Streamlit web app where users can upload images/videos and view predictions with visual explanations.

SKILLS

Programming: Python, C++, SQL
Machine Learning: PyTorch, Hugging Face Transformers, NumPy, Pandas, Scikit-learn, OpenCV
Research Areas: Membership Inference Attacks, AI Safety, ML Privacy, ML Fairness, Representation Analysis
Computer Science: Data Structures, Operating Systems, DBMS, Computer Organization and Architecture
Tools: Git, GitHub, Linux, VS Code, Google Colab, Jupyter Notebook, LaTeX, Streamlit

CERTIFICATIONS

Python Certification – C3iHub, IIT Kanpur — January 2025
Deep Learning Specialization – Andrew Ng, DeepLearning.AI — August 2025
Mathematics for Machine Learning – Imperial College London — February 2026

LEADERSHIP & ACTIVITIES

Core Member, CyberCisco — Lucknow, India
MSME-Registered Cybersecurity Initiative — September 2025 – Present

- Contributed to organizing cybersecurity webinars and coordinating technical sessions with invited speakers.